

The Beagle Conjecture: Helical (Kelvin) Packing as Universal Solution to Erdos r_k
From atoms to DNA coiling to smoke rings to tornadoes to pulsars, and everything else.
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Sophie Beagle’s definition of “helical packing”/“Kelvin packing”

$$H_{R,\omega}(\gamma)(s) = \gamma(s) + R\cos(\omega s) \hat{n}_1(s) + R\sin(\omega s) \hat{n}_2(s)$$

Iterated:

$$\gamma_{n+1} = H_{R_n, \omega_n}(\gamma_n)$$

In plain English:

γ

- ** (gamma)** — the line you start with. Could be straight wire, could be already a little helix. Think garden hose laid out.

s

- — how far along the hose you are. Like measuring tape:

$$s = 0$$

meters in.

$$s = 10$$

at start,

ten

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$$\gamma(s)$$

- — the position of hose at that measuring point. Just a dot in space.

$$\hat{n}_1(s), \hat{n}_2(s)$$

- — the two directions that stick OUT sideways from hose at that point. If hose goes forward,

$$\hat{t}$$

these are left/right and up/down. Together with forward direction , they are a little tripod that travels with you. Frenet frame, but just "sideways sticks".

$$R$$

- — radius of the new little coil you add. How big the curls are. You want

$$R$$

it small so you don't hit yourself. In DNA, is 11nm nucleosome vs 2nm

$$R$$

double helix. In your pulsar, is thickness of jet vs torus.

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ω

ω

- ** (omega)** — frequency, how many curls per meter. Big =

ω

ω

many tight curls. Small = loose lazy coil. is speed of twisting.

$\cos(\omega s)$

- ** and

$\sin(\omega s)$

** — the circle. As you walk along

s

, you go round and round in the sideways plane. Cos = side-to-side, sin =

R

up-down, together they draw circle of radius . That's the coil.

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$$H_{R,\omega}(\gamma)$$

• — new hose after you coiled it. Old
hose + circle.

$$\gamma_{n+1} = H_{R_n,\omega_n}(\gamma_n)$$

•

$$R$$

— screw of screw. Take coiled hose, coil it again with smaller and

$$\omega$$

bigger . Then again. That's DNA: helix → superhelix → looped scaffold →

$$18^\circ$$

chromosome. That's also why Crick got mismatch. tilt: 3.6 vs 3.5 heptad

The magic numbers:

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• L_n length:

$$L_{n+1} = L_n \sqrt{1 + R_n^2 \omega_n^2}$$

. Length blows up to infinity even though you stay inside same box. You pack more hose.

• $D = \log \omega / -\log R$

$$D = 1$$

fractal dimension: goes to 2. Not line () like

$$D = 2$$

straight wire, not surface (would be solid sheet).

$$1.999\dots$$

— almost surface, but still line that never touches itself. That's why it can pack infinite length.

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$$\eta = L \pi r^2 / |V|$$

.

efficiency:

$$r$$

how much of box you fill with tube thickness

. Sphere packing max 74%.

$$R \rightarrow 0$$

Helical packing $\rightarrow$ 100% as

,

$$\omega \rightarrow \infty$$

while staying self-avoiding via lift to

$$V \times \mathbb{T}^2$$

(extra hidden torus that remembers twist).

In one sentence: Take a line, add tiny circles everywhere, circles get tinier and faster each time you repeat, you get infinite whirlpool in finite box that never hits itself — that's photon,

$$r_4$$

electron, DNA, pulsar, and why

avoids 4-term progressions.

I further hypothesize, based on the construction, this is true for every k .